

WARNING: No results for query terms arnD, arnE, gtrB, kdgM, kdgN, lptC, lptF, lptG, mgrB, nanC, rfc, sigE, sopF, tagH, tagK, tisB, waaB, wbaU, wbaV!

Reference:

End Host & Metadata
Resource

Col

An Infection-Relevant Transcriptomic Compendium for *Salmonella enterica* Serovar Typhimurium
Genotype: LT2 (ATCC 14026) (bioRxiv 2019.04.01.300000) (Data Release)

[PDF \(4.6 MB\)](#), [supplement \(2 MB\)](#)

Welcome to **SalCom**, the HintonLab
Salmonella enterica serovar Typhimurium Gene Expression Compendium

Instructions: Search by gene, locus or location (multiple entries separated by comma or space, use "" for wild-card).
*To start: enter query or select **Example: *ssaG* • *STM4509* • *prgH* *ssaG* *mgIC* • *molB* *g* genes • *SPH1* • *SPH2* • *SPH1* & 2 effectors • *SPH1* • *SPH4* • *SPH5* • *SPH6* • *SPH8* • *SPH11* • *sdh* O-antigen locus • *SPH13* • *SPH14* • *SPH16* • *gtrA* locus**
arnC, arnD, arnE, bssR, csqD, flrR, gtrB, gtrB, hcp, lagB, invF, invH, kdgM, kdgN, nanC, ompL, lptC, lptF, lptG, lpxA, mgrB, n

Quality

Columns of the Heat Maps can be sorted by value - just click on the headers
(30 ~ " rotated)

Heat map for expression of 56 IDs in Kroeger *et al.*, 2013

Absolute Expression Levels (transcripts per million)

query	EEP	MEP	LEP	ESP	LSP	25°C	Cold shock (15°C)	pH5 shock	pH8.5 shock	NaCl shock	Bile shock	Low Fe ²⁺ shock	Anaerobic shock	Anaerobic growth	NonSpp2	Insp2	Peroxide shock (H ₂ O ₂)	Nitric oxide shock (N ₂ O ₂)	Common gene name	Gene Synonym	Type	Start position (474)	End position (474)	Strand	Product (EMBL DB)	4-74 ID	SL1344 Locus ID		
1 <i>lpxA</i>	313	221	201	98	10	254	188	62	222	104	272	182	76	80	182	212	141	92	19	46	-	266991	267779	+	Acyl-[acyl-carrier-protein]--UDP-N-acetylglucosamine O-acyltransferase	STM474_0237	SL1344_0229		
2 <i>pagP</i>	274	163	110	60	13	97	122	139	348	594	227	101	75	81	79	180	475	500	265	210	-	691643	692215	+	antimicrobial peptide resistance and lipid A acylation protein	STM474_0649	SL1344_0616		
3 <i>mrdB</i>	71	58	49	20	4	68	101	43	47	26	61	51	22	19	50	49	29	33	5	21	-	700318	701430	+	rod shape-determining protein RodA	STM474_0660	SL1344_0627		
4 <i>ompX</i>	1128	881	427	378	330	1443	812	546	1167	2077	1674	867	550	621	1236	685	2373	2379	1117	455	+	CDS	898795	899310	+	outer membrane protein x precursor	STM474_0858	SL1344_0808	
5 <i>bssR</i>	8	11	19	77	248	7	6	10	12	22	18	13	167	142	40	56	126	594	78	163	-	CDS	924346	924729	+	conserved hypothetical protein	STM474_0879	SL1344_0829	
6 <i>hcp</i>	9	8	7	4	3	9	5	4	6	4	7	9	6	6	9	9	5	4	1	13	-	CDS	969500	971152	+	hydroxylamine reductase, hybrid cluster [2Fe-2S] protein	STM474_0924	SL1344_0876	
7 <i>sopD2</i>	15	17	22	75	11	15	23	33	15	107	53	37	62	36	28	38	803	1267	212	376	+	CDS	1010628	1011587	+	type III secretion system effector protein	STM474_0958	SL1344_0909	
8 <i>pipA</i>	18	17	19	8	5	19	45	21	17	17	11	20	17	24	19	23	65	60	13	35	-	CDS	1131870	1132550	-	conserved hypothetical protein	STM474_1079	SL1344_1026	
9 <i>pipB</i>	1	1	7	32	3	2	4	2	2	6	3	1	2	3	9	3	679	645	57	59	-	CDS	1132772	1133647	-	Type III secretion system effector protein, homologous to pipB2, required for colonisation in chicks and for inducing the inflammatory response in the bovine ligated ileal loop model of infection	STM474_1080	SL1344_1027	
10 <i>csqD</i>	1	1	2	5	2	1	1	2	1	2	2	3	1	1	3	11	8	6	1	24	-	CDS	1185868	1186518	-	csq operon transcriptional regulator protein	STM474_1137	SL1344_1079	
11 <i>phoQ</i>	107	127	126	147	9	66	119	83	212	130	152	119	51	106	172	180	278	332	93	97	-	CDS	1274079	1275542	-	sensor protein PhoQ, regulator of virulence determinants	STM474_1228	SL1344_1168	
12 <i>phoP</i>	203	220	278	298	33	138	174	102	306	363	384	201	141	287	362	360	564	786	274	166	-	CDS	1275542	1276216	-	transcriptional regulatory protein PhoP, regulator of virulence determinants	STM474_1229	SL1344_1169	
13 <i>spiA</i>	2	1	2	7	6	1	2	1	1	7	1	1	0	1	4	2	364	242	7	40	+	CDS	1437254	1438747	+	outer membrane secretory protein	STM474_1400	SL1344_1328	
14 <i>ssaD</i>	3	1	4	9	5	3	2	1	2	8	2	3	2	2	5	7	439	229	6	33	+	CDS	1438728	1439939	+	hypothetical pathogenicity island protein	STM474_1401	SL1344_1329	
15 <i>ssaE</i>	8	4	6	103	30	5	9	6	7	26	10	17	5	15	13	16	2437	1803	2086	453	+	CDS	1439947	1440189	+	type III secretion system protein	STM474_1402	SL1344_1330	
16 <i>sseE</i>	3	3	6	17	3	4	3	1	2	4	7	2	1	2	8	8	567	528	20	67	+	CDS	1443909	1444325	+	Type III secretion system protein	STM474_1408	SL1344_1336	
17 <i>sscB</i>	7	6	9	17	13	12	10	7	7	8	5	8	5	5	9	12	604	912	22	126	+	CDS	1444377	1444811	+	Type III secretion system chaperone protein	STM474_1409	SL1344_1337	
18 <i>ssaG</i>	6	4	10	80	13	3	4	4	4	6	4	7	6	12	10	10	3244	2587	1581	620	+	CDS	1446389	1446604	+	Type III secretion system apparatus	STM474_1412	SL1344_1340	
19 <i>ssaI</i>	4	0	2	34	2	0	3	3	1	3	3	2	2	4	3	7	2147	1837	341	337	+	CDS	1446884	1447132	+	Type III secretion system apparatus	STM474_1414	SL1344_1342	
20 <i>ssaK</i>	2	2	2	23	2	1	2	1	0	3	1	1	1	2	4	4	773	736	36	143	+	CDS	1448441	1449115	+	Type III secretion system apparatus	STM474_1417	SL1344_1345	
21 <i>ssaL</i>	11	6	7	21	7	6	10	4	7	10	6	7	8	10	13	11	753	568	26	113	+	CDS	1449081	1450097	+	Type III secretion system apparatus	STM474_1418	SL1344_1346	
22 <i>ssaM</i>	8	2	4	18	5	4	6	0	1	3	2	3	4	1	5	6	1121	964	98	153	+	CDS	1450155	1450523	+	Type III secretion system apparatus	STM474_1419	SL1344_1347	
23 <i>ssaR</i>	8	5	6	22	3	3	8	5	2	4	5	6	3	4	8	7	1134	1303	239	155	+	CDS	1455596	1456243	+	Type III secretion system protein	STM474_1425	SL1344_1353	
24 <i>ssaS</i>	2	2	3	7	0	2	7	1	1	3	3	3	0	2	2	3	486	484	9	65	+	CDS	1456240	1456506	+	Type III secretion system apparatus	STM474_1426	SL1344_1354	
25 <i>ssaT</i>	4	4	3	8	2	2	7	4	2	6	4	3	2	1	5	5	596	543	5	107	+	CDS	1456507	1457286	+	Type III secretion system apparatus	STM474_1427	SL1344_1355	
26 <i>ssaU</i>	4	3	3	7	13	2	4	2	2	16	5	5	2	2	6	4	218	163	4	39	+	CDS	1457283	1458341	+	Type III secretion system protein	STM474_1428	SL1344_1356	
27 <i>steB</i>	6	4	36	132	12	4	11	3	11	19	5	3	2	29	41	17	31	31	6	19	+	CDS	1676474	1676875	+	Type III secretion system effector protein	SL1344_1559		
28 <i>sseJ</i>	1	1	1	3	3	1	1	1	1	6	0	1	1	1	1	2	532	667	180	75	+	CDS	1678146	1679372	+	Type III secretion system effector protein (SseJ)-antagonises the activity of SifA	STM474_1643	SL1344_1561	
29 <i>sopE2</i>	3	3	110	477	3	4	8	1	3	7	2	3	3	86	246	25	3	12	3	13	-	CDS	1909580	1910302	-	Type III secretion system effector protein-causes membrane ruffling and disrupts tight junctions	STM474_1879	SL1344_1784	
30 <i>fliR</i>	40	44	28	19	1	42	96	37	23	4	28	32	6	13	31	30	7	9	2	5	+	CDS	2023919	2024713	+	flagellar biosynthetic protein FlrR	STM474_2013	SL1344_1910	
31 <i>rfbN</i>	323	254	190	93	4	317	329	337	231	226	314	250	71	79	131	192	87	70	39	33	-	CDS	2163008	2163952	-	rhamnosyltransferase	STM474_2170	SL1344_2062	
32 <i>rfbU</i>	308	256	208	99	8	332	398	462	251	239	403	278	80	98	124	181	62	76	48	37	-	CDS	2163953	2165014	-	glycosyltransferase	STM474_2171	SL1344_2063	
33 <i>rfbV</i>	488	373	304	131	12	459	567	583	395	426	634	459	124	168	193	395	138	147	88	83	-	CDS	2165334	2166335	-	glycosyltransferase	STM474_2172	SL1344_2064	
34 <i>rfbG</i>	667	715	315	160	9	558	572	463	446	335	600	519	134	143	288	376	180	112	42	91	-	CDS	2169981	2171060	-	CDP-glucose 4,6-dehydratase	STM474_2176	SL1344_2068	
35 <i>sseL</i>	4	3	5	22	56	3	10	4	4	76	6	5	25	14	9	9	1036	1068	237	280	+	CDS	2392511	2393464	+	Type III secretion system effector protein, deubiquitinase	STM474_2383	SL1344_2256	
36 <i>sopE</i>	9	9	410	1838	12	17	7	14	9	10	7	5	7	323	676	58	3	16	6	53	+	CDS	2863004	2863726	+	Type III secretion system effector protein, invasion-associated secreted protein-by rearranging the actin cytoskeleton and disrupting tight junctions	STM474_2826	SL1344_2674	
37 <i>prgI</i>	22	12	49	99	5	10	8	5	8	10	16	8	4	371	1714	73	5	6	5	19	-	CDS	3038651	3038893	-	type III secretion system apparatus	STM474_3012	SL1344_2853	
38 <i>iagB</i>	14	14	205	451	46	7	11	14	17	25	17	17	24	447	628	30	6	14	10	26	+	CDS	3044111	3044593	+	cell invasion protein	STM474_3016	SL1344_2857	
39 <i>sipD</i>	8	9	176	612	14	6	7	3	5	8	7	6	5	137	567	47	4	10	5	16	-	CDS	3049331	3050362	-	pathogenicity island 1 Type III secretion system apparatus-part of the Translocon	STM474_3022	SL1344_2862	
40 <i>spaN</i>	5	4	144	305	1	3	4	1	3	2	4	3	2	141	565	24	1	3	2	3	-	CDS	3057822	3058832	-	surface presentation of antigens protein (associated with type III secretion and virulence)	STM474_3032	SL1344_2871	
41 <i>spaM</i>	9	6	216	374	1	6	2	3	3	7	2	6	7	3	240	905	38	3	4	0	2	-	CDS	3058832	3059275	-	type III secretion system secretory apparatus	STM474_3033	SL1344_2872
42 <i>invF</i>	9	8	289	510	278	6	5	4	11	4	3	4	5	312	834	36	2	5	10	18	-	CDS	3065858	3066508	-	AraC-family regulatory protein	STM474_3039	SL1344_2878	
43 <i>invH</i>	33	25	116	162	0	14	18	31	22	54	12	31																	



Relative Expression Levels (fold-change)															Relative Expression Levels (fold-change)														
query	EEP	MEP	LEP	ESP	LSP	MEP	Cold shock (15°C)	pH3 shock	pH5.8 shock	NaCl shock	Bile shock	Low Fe2+ shock	Anaerobic shock	Anaerobic growth	INSp12	INSp12 LowMg2+	Peroxide shock (INSp2)	Nitric oxide shock (INSp2)	NonSp12	INSp12	Common	Gene	Start	End	Strand	Product (EMBL DB)	4-74 ID	SL1344	
																					gene name	Synonym	Type	position (4/74)	position (4/74)				
1	arcB	1.04	0.89	0.72	0.24	1.01	0.81	0.78	0.99	1.61	1.02	1.14	1.09	1.05	0.57	1.49	1.05	0.57	1.49	1.05	arcB		CDS	3511909	3514245	-	aerobic respiration control sensor protein	STM474_3488	SL1344_33
2	bssR	1.05	1.94	7.73	24.8	0.95	0.95	1.17	2.11	1.7	1.28	15.81	0.28	4.71	0.62	1.29	1.24	0.62	1.29	1.24	bssR	yltH	CDS	924346	924729	+	conserved hypothetical protein	STM474_0879	SL1344_08
3	csgD	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	csgD		CDS	1185868	1186518	-	csg operon transcriptional regulator protein	STM474_1137	SL1344_10
4	flhR	1.11	0.72	0.47	0.25	2.18	0.85	0.52	0.23	0.64	0.73	0.23	2.5	1	1	1	1	1	1	1	flhR		CDS	2023919	2024713	+	flagellar biosynthetic protein FlhR	STM474_2013	SL1344_19
5	hcp	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	hcp	nipB	CDS	969500	971152	-	hydroxylamine reductase, hybrid cluster [2Fe-2S] protein	STM474_0924	SL1344_08
6	iagB	0.99	14.4	31.27	3.26	0.76	0.99	1.2	1.81	1.24	1.19	1.89	1.41	1.42	1	2.82	1	1	1	1	iagB		CDS	3044111	3044593	+	cell invasion protein	STM474_3016	SL1344_28
7	invF	1	28.85	50.99	27.78	1	1	1	1.08	1	1	1	2.68	1	1	1	1.83	1	1	1	invF		CDS	3065858	3066508	-	AraC-family regulatory protein	STM474_3039	SL1344_28
8	invH	0.74	3.5	4.91	0.3	0.74	1.26	0.89	2.19	0.5	1.25	0.76	2.28	1	1	1	1.61	1	1	1	invH		CDS	3066965	3067408	+	outer membrane lipoprotein	STM474_3040	SL1344_28
9	ompL	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	ompL	ysaH	CDS	4246449	4247141	-	conserved hypothetical protein	STM474_4193	SL1344_39
10	lpxA	0.71	0.64	0.31	0.03	0.86	0.28	1.01	0.47	1.23	0.83	0.34	2.26	0.65	0.14	0.33	1	0.66	1	0.66	lpxA		CDS	266991	267779	+	Acyl-[acyl-carrier-protein]-UDP-N-acetylglucosamine O-acetyltransferase	STM474_0237	SL1344_02
11	mrcA	0.76	0.77	0.34	0.21	1.38	0.65	1.01	0.34	1.08	0.87	0.5	1.89	0.65	0.33	0.48	1	0.71	1	0.71	mrcA		CDS	3668148	3670724	+	penicillin-binding protein 1A	STM474_3660	SL1344_34
12	mrdB	0.82	0.69	0.28	0.14	1.73	0.74	0.8	0.44	1.05	0.87	0.38	2.67	1	1	0.34	0.7	1	0.6	1	mrdB		CDS	700318	701430	-	rod shape-determining protein RodA	STM474_0660	SL1344_06
13	ompR	1.05	1.13	0.91	0.12	1.1	0.44	1.49	1.03	1.65	1.14	0.95	1.57	0.91	0.45	0.43	1	2	1	2	ompR		CDS	3681128	3681847	-	two-component response regulator OmpR	STM474_3669	SL1344_34
14	ompX	0.79	0.38	0.33	0.29	0.91	0.61	1.31	2.33	1.88	0.97	0.62	1.99	1	1.05	0.49	0.2	1	3.32	1	ompX		CDS	896795	899310	+	outer membrane protein x precursor	STM474_0858	SL1344_08
15	pagP	0.6	0.4	0.22	0.05	0.74	0.85	2.13	3.64	1.39	0.62	0.46	0.98	1	1.05														

query	EEF	MEP	LEP	ESP	LSP	MEP	Cold shock (15°C)	pH5 shock	pH5.8 shock	NaCl shock	Bile shock	Low Fe2+ shock	Anaerobic shock	Anaerobic growth	Oxygen shock	InsP12	InsP12 LowMg2+	Peroxide shock (InsP12)	Nitric oxide shock (InsP12)	NonSP12	Common gene name	Gene Synonym	Type	Start position (4/74)	End position (4/74)	Strand	Product (EMBL DB)	4-74 ID	SL1344 Locus ID	
	1	1	1	7.99	1.31	1	1	1	1	1	1	1	1	0.87	1	1.11	0.33	0.19	1	319.02									ssaG	CDS
43 ssaG	1	1	1	7.99	1.31	1	1	1	1	1	1	1	1	1	1	1	1	0.86	0.16	0.16	1	214.71	ssaI	CDS	1446884	1447132	+	Type III secretion system apparatus	STM474_1414	SL1344_13
44 ssaI	1	1	1	3.41	1	1	1	1	1	1	1	1	1	1	1	1	1	0.95	0.05	0.19	1	77.29	ssaK	CDS	1448441	1449115	+	Type III secretion system apparatus	STM474_1417	SL1344_13
45 ssaK	1	1	1	2.33	1	1	1	1	1	1	1	1	1	1	1	1	1	0.75	0.03	0.15	1	66.99	ssaL	CDS	1449081	1450097	+	Type III secretion system apparatus	STM474_1418	SL1344_13
46 ssaL	1	0.91	0.91	1.88	0.91	1	1.02	1	1	1.02	1	1	1	1	1	1	1	0.86	0.05	0.14	1	112.11	ssaM	CDS	1450155	1450523	+	Type III secretion system apparatus	STM474_1419	SL1344_13
47 ssaM	1	1	1	1.75	1	1	1	1	1	1	1	1	1	1	1	1	1	1.15	0.21	0.14	1	113.43	ssaR	CDS	1455596	1456243	+	Type III secretion system protein	STM474_1425	SL1344_13
48 ssaR	1	1	1	2.19	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0.02	0.13	1	48.61	ssaS	CDS	1456240	1456506	+	Type III secretion system apparatus	STM474_1426	SL1344_13
49 ssaS	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1.06	0.02	0.18	1	69.64	ssaT	CDS	1456507	1457286	+	Type III secretion system apparatus	STM474_1427	SL1344_13
50 ssaT	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0.75	0.05	0.18	1	21.81	ssaU	CDS	1457283	1458341	+	Type III secretion system protein	STM474_1428	SL1344_13
51 ssaU	1	1	1	1	1.32	1	1	1	1	1.63	1	1	1	1	1	1	1	1.52	0.04	0.21	1	51.88	sscB	CDS	1444377	1444811	+	Type III secretion system chaperone protein	STM474_1409	SL1344_13
52 sscB	1	1	1	1.71	1.34	1	1.05	1	1	1	1	1	1	1	1	1	1	0.93	0.03	0.12	1	66.74	sseE	CDS	1443909	1444325	+	Type III secretion system protein	STM474_1408	SL1344_13
53 sseE	1	1	1	1.73	1	1	1	1	1	1	1	1	1	1	1	1	1	1.25	0.34	0.14	1	53.24	sseJ	CDS	1678146	1679372	+	Type III secretion system effector protein (SseJ)-antagonises the activity of SifA	STM474_1643	SL1344_15
54 sseJ	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1.5	0.23	0.27	1	193.68	sseL	CDS	2392511	2393464	+	Type III secretion system effector protein, deubiquitinase	STM474_2383	SL1344_22
55 sseL	1	1	1	2.22	5.55	1	1	1	1	7.64	1	1	2.52	1	0.73	1	1.5	0.23	0.27	1	193.68	sseL	CDS	2392511	2393464	+	Type III secretion system effector protein, deubiquitinase	STM474_2383	SL1344_22	
56 steB	1	1	3.58	13.16	1.21	1	1.08	1	1.11	1.87	1	1	1	1	1	1	1	0.33	0.63	1	1.86	steB	CDS	1676474	1676875	+	Type III secretion system effector protein		SL1344_15	

SCALE

Down-Regulated >6-foldNo ChangeUp-Regulated >6-fold

download data: 